

Lecture 2

Theory of demand and supply analysis

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The **desire**,
ability, and
willingness
to buy a
product or
service

Desire? Ability? Willingness?



Desire

Desire is **just a wish** on the part of the consumer to possess a commodity.

Want

If the desire to take a commodity is backed by the **purchasing power** and the consumer is also **willing to buy that commodity**, it becomes want.

Demand

Demand is the wish of the consumer to get a **definite quantity of a commodity** at a **given price in the market** backed by a sufficient purchasing power.

There are **three important points** to remember about the **quantity demanded**:

- **First**, the quantity demanded is the **quantity desired to be purchased**. It is the **desired purchase**. The quantity **actually bought** is referred to as **actual purchase**.
- **Secondly**, quantity demanded is always considered as a **flow measured over a period of time**.
 - if the quantity demanded of oranges is 10, it must be per day or per week, etc.
- **Thirdly**, the quantity demanded will have an **economic meaning only at a given price**.
 - For example, the demand for oranges equal to 10 units per week at a price of Rs. 100 per **dozen** is a full and meaningful statement, as used in micro-economic theory.

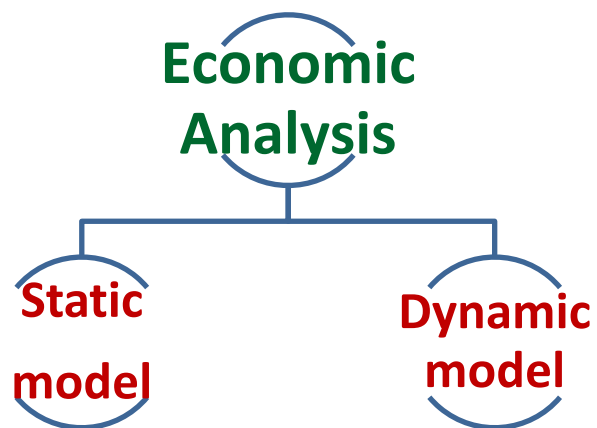
Economic Analysis

Static Economics

- Situation where movement is continuous, regular and const. i.e. there are no **unexpected** change.
- Economic conditions are assumed to be given
- Related with particular point of time hence **called timeless economy.**
- It studies only **one particular point** of equilibrium.

Dynamic Economics

- It deals with realistic condition where economy is in motion and unexpected changes (new taxes, subsidy) may happen.
- It is close to **real world economic condition.**
- Economic conditions continuous to change over time.
- Related with particular period of time. Time plays important role hence all quantities **must be dated.**



Static Model

- Variables are not dated
- The demand-supply model of market behaviour is a static model.
- In this model, **demand depends on own price**, supply depends on **own price**, with an equilibrium condition that demand must equal supply, time does not enter into the picture at all and the variables are all undated.

Dynamic Model

- Variables are dated
- If the demand-supply model is restructured as follows, then the model would become dynamic according to this criterion:

$$D_t = f(P_t)$$

$$S_t = g(P_t)$$

$$D_t = S_t$$

- D_t : demand at time t
- S_t : supply at time t
- P_t : price at time t

The first two equations say:

- Demand at time t depends on current price
- Supply at time t depends on current price

The third equation is the market-clearing condition

- Demand equals supply in the same period

- Where, ' t ' is the relevant time unit.

- But According to some economists, even if the variables are dated the model does not become dynamic.

- According to this definition, variables must be dated and a time lag must exist in their relationships. **With this criterion, a dynamic model would be:**

$$\begin{aligned}D_t &= f(P_t) \\ S_t &= g(P_{t-1}) \\ D_t &= S_t\end{aligned}$$

- Today's supply depends on **yesterday's price**
- The system evolves over time → **dynamic**

The price level in previous period (t-1) would have induced the producers to increase or decrease the supply, full impact of such decisions are visible in time period 't' only.

- ✓ The demand for commodity is dependent on a number of factors. These are mentioned as follows:

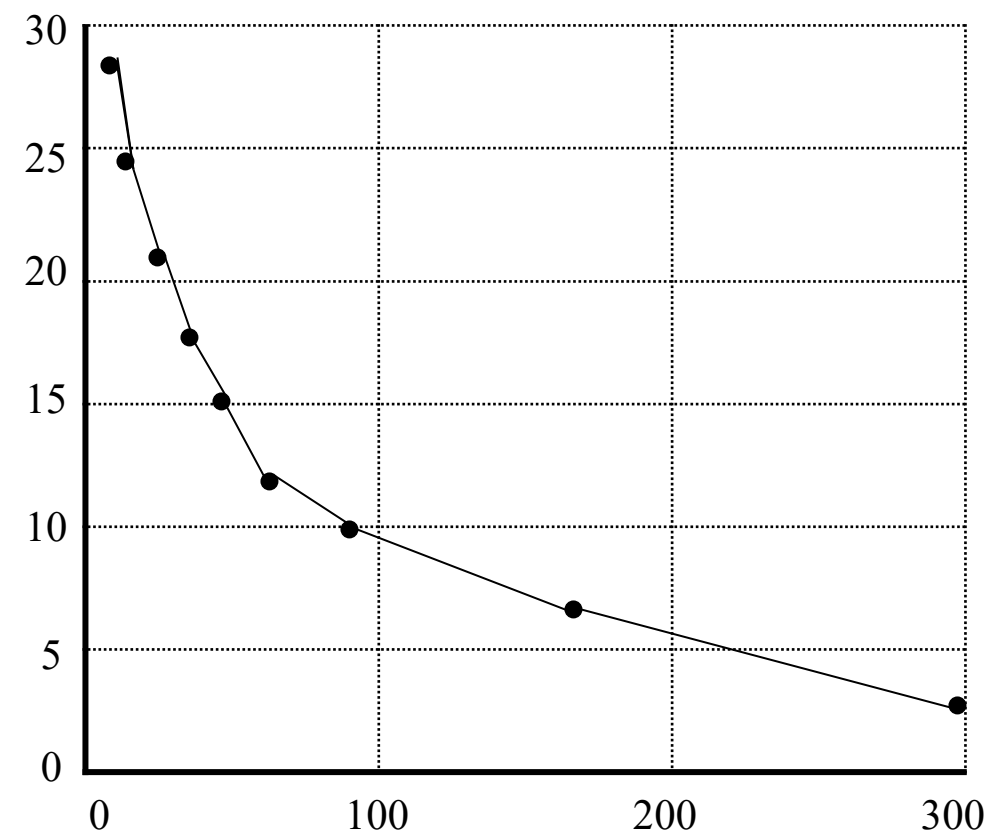
1

Price of the commodity: Normally, higher the price of the commodity, the lower the demand of the commodity. This is the **law of demand**.

Price Per CD	# of CDs Demanded
\$1	300
\$2	162
\$3	94
\$4	58
\$5	37
\$6	25
\$10	18
\$15	13
\$20	10

High price → low quantity demanded

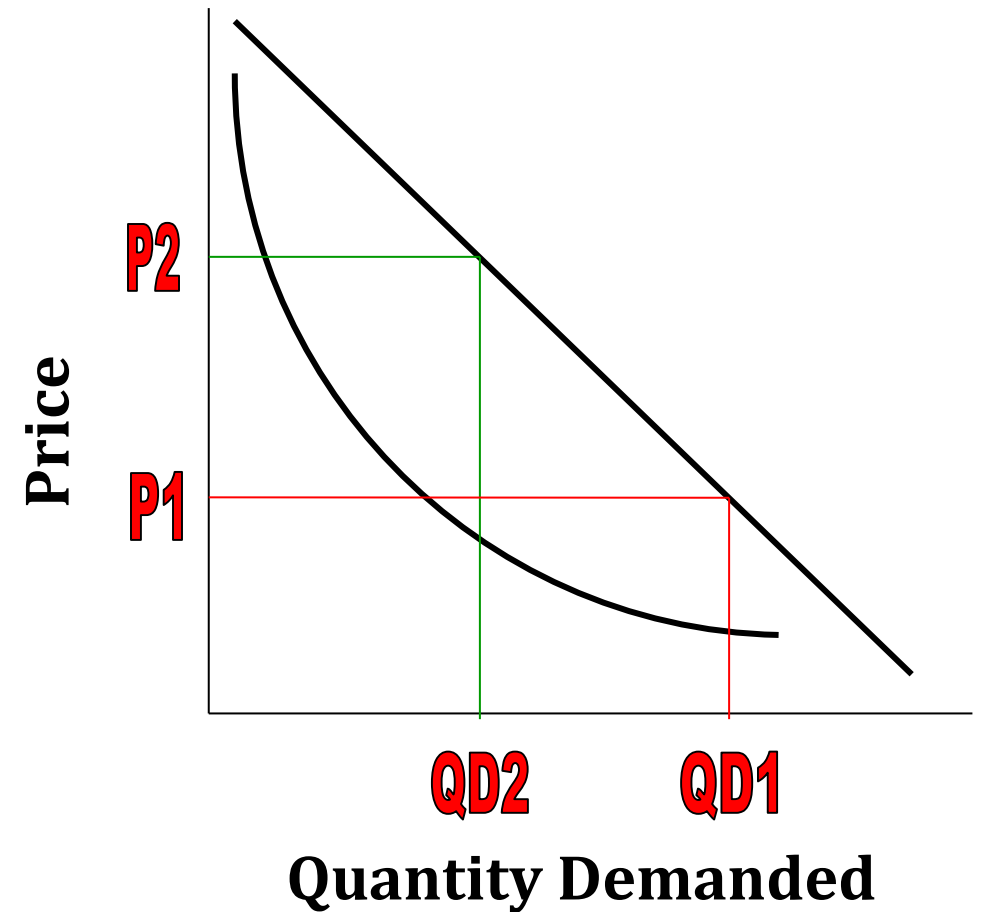
Low price → high quantity demanded



Law of demand.

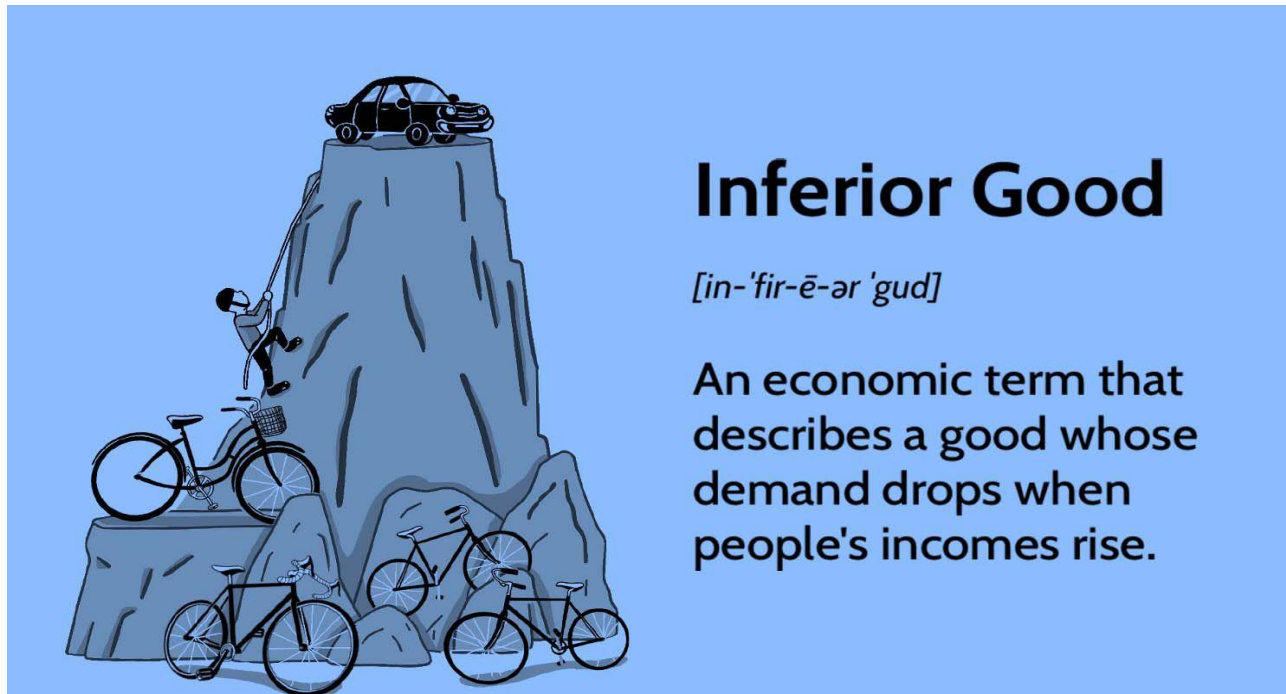
P= Price QD= Quantity Demanded

$P \uparrow$ $QD \downarrow$
 $P \downarrow$ $QD \uparrow$



2

- Size of consumers income/ Buyer's income: When the increase in income leads to an increase in the quantity demanded, the commodity is called a **'normal good'**. If an increase in income leads to a fall in the quantity demanded, we call that commodity an **'inferior good'**.



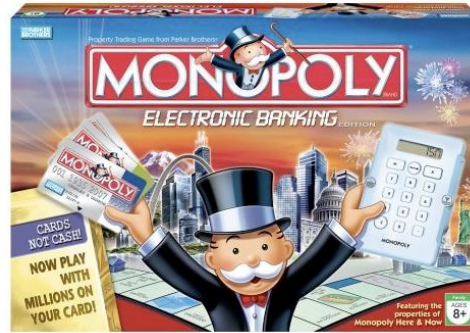
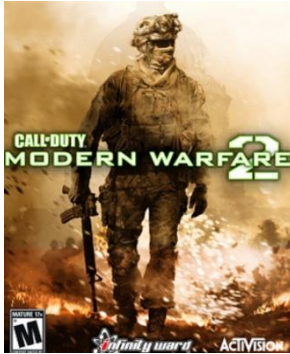
3

Prices of other related commodities: A consumer's demand for a commodity may also be influenced by the prices of some other commodities. Some are **complementary goods**, which are consumed along with the commodity, while others may be used in place of this commodity. This category is called **substitutes**.



4

Taste of the consumers: If a consumer has **developed a taste** for a particular commodity, he/she will demand more of that commodity. Similarly, if a consumer has **changed his taste** against a particular commodity, less of it will be demanded at any particular price. This development of tastes may be related to seasons of the year as well.



- Demand function refers to the rule that shows how the quantity demanded depends upon above factors.
- A demand function can be shown as:

$$D_x = f(P_x, P_y, P_z, M, T)$$

where,

D_x is quantity demanded of X commodity

P_x is the price of X commodity

P_y is the price of substitute commodity

P_z is price of a complement good

M stands for income

T is the taste of the consumer

➤ If all the factors influencing the demand for a commodity X vary simultaneously, **the picture would be highly complicated.**

➤ Therefore, normally we allow only one of the factors to change, **assuming that all other factors remain unchanged**

- ✓ The factors determining the demand for a commodity in a market are the same as those which determine the demand for the commodity on the part of a consumer.
- ✓ Besides that two additional factors are also to be included. These two factors are:

- **Size of the population:** All other factors remaining unchanged, the greater is the size of the population, more of a commodity will be demanded.

- **Income distribution:** People in different income groups show marked differences in their preferences. So if larger share out of national income goes to the rich, demand for the luxury goods may rise and a rise in income share of the poor will increase demand for the wage goods.

Engineering economics deals with the methods that enable one to take economic decisions towards minimizing costs and/or maximizing benefits to business organizations.

Key points

- Example: Technical efficiency of a diesel engine is less than 100%.
- ✓ This is mainly due to frictional loss and incomplete combustion of fuel
 - Technical efficiency can never be more than 100%.

- Worth: Annual revenue generated by way of operating the business.
- Cost: Total annual expenses incurred in carrying out the business.
 - For the survival and growth of any business, the economic efficiency should be more than 100%.

